

Nandhakumar V

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AI / ML ENGINEER — JUNIOR

AI & Data Science graduate with verified project experience in machine learning, deep learning and NLP using Python, Scikit-learn, TensorFlow/Keras and Transformers. Suitable for junior/entry-level AI/ML roles where project experience is accepted; no unverified production LLM/RAG/agent claims are included.

Education

Dhirajlal Gandhi College Of Technology 2021 – 2025
B.Tech – Artificial Intelligence and Data Science | CGPA: 7.8

Work History

Data Science Consultant, Intern | Rubixe – AI Solutions Bengaluru (Onsite) | Jun 2025 – Jan 2026

- Developed and evaluated machine-learning workflows using Python and Scikit-learn for classification and regression tasks.
- Applied EDA, statistical analysis, preprocessing and feature engineering on structured datasets for data-driven analysis.
- Contributed to the IT Incident Management client project using approximately 46K enterprise records for incident prioritisation.
- Completed the Certified Data Scientist (CDS) programme through the DataMites learning pathway alongside project and practical work.

Full Stack Developer | Aggregate Intelligence Inc Coimbatore (Onsite) | Aug 2024 – Jan 2025

- Worked onsite on full-stack app development, contributing to backend integration and application workflows.
- Developed Python backend components and REST APIs for application and data-integration workflows.
- Worked with MySQL/database operations and SQL query optimisation, with approximately 20% lower query latency.
- Used Git/GitHub and Docker in application development, testing and integration workflows.

Data Science Course & Projects | EdiGlobe Pvt Ltd Bengaluru (Virtual) | Aug 2024 – Oct 2024

- Completed structured self-paced Data Science learning and project work using Python and data-analysis libraries.
- Performed data cleaning, EDA, statistical analysis and visualization using Pandas, NumPy, Matplotlib and Seaborn.
- Applied preprocessing and feature engineering to prepare datasets for predictive modelling and analysis.
- Completed House Price Prediction and IMDb Movie Data Analysis projects.

Skills

- **Programming & ML:** Python, SQL, Pandas, NumPy, Scikit-learn
- **Machine Learning:** Classification, Regression, Feature Engineering, Data Preprocessing, Model Evaluation, Logistic Regression, KNN, SVM, Decision Trees, Random Forest, XGBoost, LightGBM, CatBoost
- **Deep Learning & NLP:** TensorFlow, Keras, CNN, LSTM, ANN, Transformers, NLTK, NLP, Text Preprocessing, Sentiment Analysis, Image Classification
- **Backend & Software:** Flask, REST APIs, Git/GitHub, Docker
- **Training / Basics:** PyTorch (Basics), Prompt Engineering, LLM concepts, FastAPI (Basics)

Certifications

- **Certified Data Scientist — NASSCOM / FutureSkills Prime** | Gold Category | 91% Assessment Score | Apr 2026
- **Data Science Foundation — IABAC** | Effective May 2026
- **Certified Data Scientist (CDS-DS2050) — DataMites** | 700+ hours | IABAC / NASSCOM accredited
- **Data Science Foundations – Level 2 (V2) — IBM / Credly** | Sep 2025
- **Data Science 101 — IBM** | Sep 2025
- **Data Science Tools — IBM** | Sep 2025
- **Data Science Methodology — IBM** | Sep 2025
- **Python (Basic) — HackerRank** | Sep 2025

Projects

AI-Powered Sentiment Analysis

- Built a customer-review sentiment analysis workflow using the Amazon Datafiniti dataset containing 5,000 rows and 24 columns.
- Goal: classify customer sentiment automatically and support scalable analysis of review feedback.
- Applied text preprocessing, NLTK, TensorFlow/Keras and Transformers, with comparative modelling including LSTM and Random Forest.
- Verified results: Random Forest 94.8% accuracy and LSTM 93.7%.

Skills Applied: Python | NLP | NLTK | TensorFlow | Keras | LSTM | Transformers | Sentiment Analysis

IT Incident Management using Machine Learning

- Analysed approximately 46K enterprise ITSM records from a live MySQL database for incident prioritisation and operational analytics.
- Goal: improve ticket triage and prioritisation and reduce manual effort in IT incident management.
- Applied SQLAlchemy, data cleaning, ITIL Impact × Urgency priority logic, feature engineering, Random Forest, Prophet and Gradient Boosting with SMOTE across separate workflows.
- Verified results included High-Priority Random Forest F1-score of approximately 0.98, Auto-Tagging around 83% and RFC-Failure around 93%.

Skills Applied: Python | SQL | MySQL | SQLAlchemy | EDA | Feature Engineering | Scikit-learn | Random Forest.

Skin Disorder Detection

- Built a multiclass classification workflow using a dermatology dataset with 366 rows, 35 features and six disorder classes.
- Goal: classify skin-disorder categories from structured clinical attributes using machine-learning methods.
- Applied preprocessing and compared Logistic Regression, KNN and Random Forest using classification metrics.
- Verified test accuracy: Random Forest 100%, Logistic Regression 98.65%, KNN 85.14% and SVM 75.68%.

Skills Applied: Python | Scikit-learn | Classification | Logistic Regression | KNN | SVM | Random Forest

Rice Leaf Disease Detection

- Built a CNN-based image-classification workflow using 119 rice-leaf images across Bacterial Blight, Brown Spot and Leaf Smut.
- Goal: demonstrate automated rice-leaf disease classification from images.
- Used 128×128 image processing, an 80/20 split, ImageDataGenerator augmentation and TensorFlow/Keras CNN model comparison.
- Verified best model accuracy was 71%.

Skills Applied: Python | TensorFlow | Keras | CNN | Image Classification | Data Augmentation

Earthquake Damage Prediction

- Developed a multiclass classification workflow for the DrivenData earthquake-damage dataset using structural and geographical features.
- Goal: predict building damage categories for data-driven post-earthquake risk assessment.
- Performed data merging, EDA, IQR-based outlier capping and OneHotEncoding; compared Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM and CatBoost.
- Evaluated model performance to identify the most suitable approach for earthquake-damage classification.

Skills Applied: Python | Pandas | Scikit-learn | XGBoost | LightGBM | CatBoost | EDA

Home Loan Default Risk Management

- Developed a machine-learning workflow using Home Credit data; application_train contains 307,511 rows, with seven CSV sources used in the broader analysis.
- Goal: identify customers at risk of loan default and support data-driven credit-risk assessment.
- Applied multi-table integration, preprocessing, memory-aware optimisation including Numba JIT, feature engineering and model comparison; SMOTE was used only for the RFC sub-model.
- Verified result: XGBoost was the strongest reported model at approximately 92% accuracy; XGB AUC was 0.760.

Skills Applied: Python | Pandas | Scikit-learn | XGBoost | Feature Engineering | Data Preprocessing Visualization